AMUTHEEZAN SIVAGNANAM

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SUMMARY

PhD Candidate and **Machine Learning Researcher** specializing in **Deep Reinforcement Learning** with **six** years of academic research and **two** years of industrial software development experience. Expertise in **scalable deep learning and reinforcement learning systems**, ML **training, and deployment**, complemented by a **strong publication record (h-index 6)**. Proficient in **Python**, **TensorFlow**, **PyTorch**, AWS, Agile and CI/CD practices, with hands-on experience in **optimizing ML models and pipelines**.

EDUCATION

Pennsylvania State University
Ph.D., Informatics — *Aug 2025*
Dissertation: *Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains*

University of Houston
M.S., Computer Science — *Aug 2022*

University of Moratuwa
B.S., (Hons) Engineering (Computer Science and Engineering) — *Jan 2018*
Final Year Project: *Sentimental Analysis of Twitter using Semi-Supervised Approaches*

AREAS OF EXPERTISE

Reinforcement Learning, Optimization, Operational Research

SKILLS

Programming Languages: Python, C++, Java, C, JavaScript, PHP, MATLAB
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Ray, RLLib, OpenAI Gym,

NumPy, pandas, matplotlib, CPLEX, Gurobi, Mosek, OR-Tools
Databases: SQL, MySQL, SQLite, MongoDB
Tools & Platforms: Amazon Web Services, Git, GitHub, Linux, Docker, Spark, CI/CD, Bash
Methodologies: Object-Oriented Development, Agile, Behaviour Driven Development

RESEARCH EXPERIENCE

Pennsylvania State University — Graduate Research Assistant
Applied Artificial Intelligence Lab
Aug. 2022 - May 2025

- Lead **research projects** funded by the **U.S. Department of Energy (DOE)** and the **National Science Foundation (NSF)**, introducing **cost- and energy-efficient solutions** to tackle real-world problems
- **Analyzed** current state-of-the-art solutions and identified **research gaps** in existing solution approaches, leading to the development of **innovative strategies** for **real-time decision-making**
- Introduce novel **problem formulations and mathematical models** to address these challenges, while ensuring inherent real-world constraints
- Proposed **artificial intelligence–based solutions** to tackle **real-world problems** and successfully **deployed** them in relevant **industries**
- Published **research findings** in **AI conferences** (**ICML**) and assisted in preparing **slides** for presenting results at **DOE** and **NSF meetings**

The University of Houston — Graduate Research Assistant
Resilient Networks and Systems Lab
Sep. 2019 - Aug. 2022

- Spearheaded **research projects** funded by the **U.S. Department of Energy (DOE)** and the **National Science Foundation (NSF)**, leading to advancements in **energy-efficient technologies**
- Focus on understanding **real-world decision-making problems** and identified **gaps** in existing solution approaches, leading to the development of **novel methodologies** for **enhanced decision-making**
- Propose **novel problem formulations** and **mathematical models** and formulated **problem statements** to effectively address the research limitations in prior research efforts
- Applied **artificial intelligence-based solution approaches** to **real-world problems** and successfully **deployed** them in relevant **industries**
- Published **research findings** in **AI conferences** (**AAAI**, **IJCAI**) and assisted preparing **slides** for presenting results at **DOE** and **NSF meetings**

SOFTWARE ENGINEERING EXPERIENCE

LSEG Technology — Software Engineer
Post Trade Team
Jan 2018 – Jul 2019

- Developed application software using **Object-Oriented principles**, enhancing **code maintainability** and **scalability**
- Implemented **agile development practices**, such as **Scrum**, to improve **team collaboration** and **project delivery speed**
- Developed software solutions using **Java**, **Python**, and **C++**, improving **system performance** and **reliability**
- Introduced **Unit Testing** for **Libraries** in **Post Trade C++ Code**
- Modified **database structures** and tested using **BDD-based testing**, ensuring **data integrity** and **system functionality**
- Identified and replaced **duplicated error codes** with valid new error codes, improving **system reliability**
- Developed a **dynamic scripting object (DSO)** for **front-end end-to-end testing** using **Java**, enhancing **test coverage** and **efficiency**
- Analyzed and modified **back-end regression scripts** to work with both old and new **testing frameworks**, ensuring **compatibility** and reducing **testing time**
- Developed a new **report generation system** for the testing framework, **automating email dispatch** at the end of regression and improving **communication efficiency**
- Completed **code integration tasks** related to **back-end regression**, streamlining the **development process** and enhancing **system performance**
- Updated **automatic updates** to **auto-generated codes** based on **Database changes** using **Integration Plans**
- Contributed to **code integration** and **deployment plans**, ensuring **seamless software updates** and minimizing **downtime**
- Participated in **professional training programs** conducted by **Millennium IT Software** and **Post Trade Team**
- Worked on **Front-End Development** for both **Product** and **Solution** which consists of **Enhancement**, **Bug Fixing**, **Merging** and **Introducing new features**

WSO2 Lanka PVT Ltd — Software Engineering Intern
Data Analytics Team
Jul 2016 – Dec 2016

- Developed an HL7 Monitoring Solution by integrating **WSO2 ESB**, **DAS**, and **BAM**, enabling real-time and batch processing of healthcare data
- Developed an alert generation system analyzing descriptive **HL7/FHIR** data to monitor **disease outbreaks**, triggering timely email and SMS notifications
- Created **Spark scripts** for batch analytics and **Siddhi execution plans** for real-time alerts, including disease outbreak and patient wait-time notifications

- Engineered a mechanism to evaluate **hospital functionality** by assessing **admission and discharge** messages, including bed and oxygen cylinder availability
- Implemented interactive dashboards using **Jaggery**, **JavaScript**, **jQuery**, **Leaflet.js**, and **DataTables**, featuring gadgets like charts, maps, and tables
- Utilized **HAPI library and test panel** to simulate HL7 v2 messages and validate the monitoring pipeline end-to-end
- Packaged the entire monitoring solution as a **Carbon Application (CApp)**, bundling event streams, receivers, publishers, and visualization artifacts
- Attended workshops and gained hands-on experience with **Git**, **MSF4J (Microservices for Java)**, and **WSO2 product architecture** for enterprise middleware solutions

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PUBLICATIONS

CONFERENCE PROCEEDINGS

[C7] **Sivagnanam, A.,** Pettet, A., Lee, H., Mukhopadhyay, A., Dubey, A., & Laszka, A. (2024). Multi-agent reinforcement learning with hierarchical coordination for emergency responder stationing. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

[C6] Sen, R., **Sivagnanam, A.**, Laszka, A., Mukhopadhyay, A., & Dubey, A. (2024). Grid-aware charging and operational optimization for mixed-fleet public transit. In *Proceedings of the 2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)* (pp. 4172–4179). IEEE.

[C5] Pavia, S., Rogers, D., **Sivagnanam, A.**, Wilbur, M., Edirimanna, D., Kim, Y., Mukhopadhyay, A., Pugliese, P., Samaranayake, S., Laszka, A., & Dubey, A. (2024). SmartTransit.AI: A dynamic paratransit and microtransit application. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI '24)* (Article No. 1028, pp. 4).

[C4] Atefi, S., **Sivagnanam, A.**, Ayman, A., Grossklags, J., & Laszka, A. (2023). The benefits of vulnerability discovery and bug bounty programs: Case studies of Chromium and Firefox. In *Proceedings of the ACM Web Conference 2023 (WWW '23)* (pp. 2209–2219). Association for Computing Machinery.

[C3] **Sivagnanam, A**., Kadir, S. U., Mukhopadhyay, A., Pugliese, P., Dubey, A., Samaranayake, S., & Laszka, A. (2022, July). Offline vehicle routing problem with online bookings: A novel problem formulation with applications to paratransit. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)* (pp. 3933–3939).

[C2] **Sivagnanam, A.**, Ayman, A., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021, May). Minimizing energy use of mixed-fleet public transit for fixed-route service. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 17, pp. 14930–14938).*

[C1] Ayman, A., Wilbur, M., **Sivagnanam, A.**, Pugliese, P., Dubey, A., & Laszka, A. (2020). Data-driven prediction of route-level energy use for mixed-vehicle transit fleets. In *Proceedings of the 2020 IEEE International Conference on Smart Computing (SMARTCOMP)* (pp. 41–48). IEEE.

JOURNALS

[J2] Wilbur, M., Ayman, A., **Sivagnanam, A.**, Ouyang, A., Poon, V., Kabir, R., Vadali, A., Pugliese, P., Freudberg, D., Laszka, A., & Dubey, A. (2023). Impact of COVID-19 on public transit accessibility and ridership. *Transportation Research Record, 2677*(4), 531–546.

[J1] Ayman, A., **Sivagnanam, A.**, Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021). Data-driven prediction and optimization of energy use for transit fleets of electric and ICE vehicles. *ACM Transactions on Internet Technology, 22*(1), Article 7, 1–29.

WORKSHOPS

[W1] **Sivagnanam, A.**, Atefi, S., Ayman, A., Grossklags, J., & Laszka, A. (2021). On the benefits of bug bounty programs: A study of Chromium vulnerabilities. In *Workshop on the Economics of Information Security (WEIS)* (Vol. 10).

BOOK CHAPTERS

[BC1] Wilbur, M., **Sivagnanam, A.,** Ayman, A., Samaranayake, S., Dubey, A., & Laszka, A. (2023, August). Artificial intelligence for smart transportation. In **Y. Vorobeychik & A. Mukhopadhyay (Eds.),** *Artificial Intelligence and Society* (book chapter). ACM Press.

PROJECTS

PHD RESEARCH PROJECTS

Emergency Responder Stationing

November 2021 – June 2024

- **Developed a novel Multi-Agent Deep Reinforcement Learning (DDPG)** framework with **hierarchical coordination** to address the **emergency responder stationing problem**

- In the hierarchical setup, **DDPG agents** manage **city-scale redistribution** (high-level) and **region-scale reallocation** (low-level)
- **Utilized a Transformer-based actor network** to handle **variable numbers of responders** in region-scale reallocation
- Ensured **feasible and exact mapping from continuous to discrete actions** using **min-cost flow** (city-level) and **max-weight matching** (region-level), while **preserving gradient flow** during training
- **Integrated low-level critics** to provide **reward feedback** to high-level agents, enhancing **training stability and performance**
- **Achieved 1000× faster decision-making** and **reduced response delays by 5–13 seconds** on **real-world datasets from Nashville and Seattle**

The Benefits of Vulnerability Discovery and Bug Bounty Programs

February 2020 – May 2023

- Collect the publicly available **Chromium data set** using Monorail API, Google Release Notes and Google Chrome Hall of Fame
- Perform intensive **data cleaning** process to **identify the original reporters**, **duplicates issues, and time at which the issue got patched and released to public**
- Convert the processed data into a **simple relational database** using **SQLite** with two tables to represent the **Issues** and **Comments** to the issues
- Conducted a comprehensive empirical analysis of **Chromium Vulnerability Reward Program (VRP)** reports, focusing on trends across stable and development versions
- Provided evidence that **vulnerabilities in stable releases are harder to detect**, validating the **security impact** of structured VRPs
- Demonstrated that **bug-bounty programs complement internal security teams** by uncovering a **diverse range of vulnerability types** beyond in-house detection capabilities
- Delivered **actionable insights** to enhance bug-bounty effectiveness, including **targeted guidance for bug hunters** toward vulnerabilities with higher real-world exploitation potential

Offline Vehicle Routing Problem with Online Bookings

February 2020 – July 2022

- Focus on a real-world paratransit operations where **riders book trips in advance with time flexibility** but expect **tight pickup time windows at booking time**
- Introduced **a novel problem formulation**: the **Offline Vehicle Routing Problem with Online Bookings**, which blends the **scalability of Offline VRP** with the **real-time responsiveness of Dynamic VRP**
- Developed a **Deep Reinforcement Learning–based policy** that learns to assign optimal time windows under demand uncertainty and booking-time constraints

- Integrated an **anytime VRP solver** to incrementally **refine and improve route plans between bookings**, enabling better long-term efficiency
- Achieved **20–40% cost reduction** over baseline methods with naive window assignments, as demonstrated through extensive **experiments on real-world paratransit of Chattanooga**

Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service

August 2019 – May 2021

- **Formulated a mathematical model** to optimize energy consumption for public transit agencies operating **mixed fleets of Electric Vehicles (EVs) and Internal Combustion Engine Vehicles (ICEVs)**
- Transformed the model into an **Integer Programming (IP) formulation** to obtain **exact optimal solutions** for **small to medium-sized** problem instances
- Developed **a scalable solution approach** combining **heuristics and metaheuristics** (e.g., **local search**, **genetic algorithms**) to **efficiently solve large-scale instances** in **polynomial time**
- **Reduced annual energy costs by $140K** for the **Chattanooga public transit agency** through the proposed approach

UNDERGRAD RESEARCH PROJECT

Sentimental Analysis of Twitter using Semi-Supervised Approaches

November 2016 – November 2017

- Developed a **sentiment analysis model** to classify tweets into **positive, negative, and neutral** categories
- Addressed **limited and noisy labeled data** by applying **semi-supervised learning techniques**, including **self-training**, **co-training**, and **lightweight ensemble methods**
- Initiated training with a **small hand-labeled dataset**, and iteratively expanded it by **automatically labeling high-confidence tweets**
- Refined the classifier through **multiple training iterations**, improving accuracy and robustness on real-world social media data

HL7 Monitoring System

July 2016 – December 2016

- **Built an end-to-end HL7/FHIR monitoring system** using WSO2 ESB, DAS, and BAM for both real-time (Siddhi) and batch (Spark) analytics of healthcare data
- **Designed an alert system** to detect disease outbreaks and long patient wait times, delivering **email and SMS notifications** based on HL7/FHIR data streams

- **Engineered hospital functionality assessments** by analyzing admission/discharge events, tracking resources like **bed and oxygen cylinder availability**
- **Developed interactive dashboards** with **Jaggery, JavaScript, Leaflet.js**, and **DataTables**, and packaged the solution as a **WSO2 Carbon Application (CApp)** for easy deployment

TALKS

IJCAI (Vienna, Austria) 2022
For the paper: Offline Vehicle Routing Problem with Online Bookings:
A Novel Problem Formulation with Applications to Paratransit

WEIS (Virtual Workshop) 2021
For the workshop paper: On the Benefits of Bug Bounty Programs:
A Study of Chromium Vulnerabilities

AAAI (Virtual Conference) 2021
For the paper: Minimizing Energy Use of Mixed-Fleet Public Transit
for Fixed-Route Service

SERVICES

Program Committee Member
International Joint Conference on Artificial Intelligence (AI For Social Good) 2025

Reviewer
International Joint Conference on Artificial Intelligence (AI For Social Good) 2024

Reviewer
AI4Research Workshop @IJCAI2024 2024

Auxiliary Reviewer
22nd International Conference on Autonomous Agents and Multiagent Systems 2023

AWARDS AND HONORS

Information Security Quiz 2015 - Runners Up 2015
Issued by Sri Lanka Computer Emergency Readiness Team(SLCERT)
and Information and Communication Technology Agency of Sri Lanka(ICTA)

Mahapola Scholarship 2014 - 2017

REFERENCES

Dr. Aron Laszka (PhD Supervisor)
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